



Presumptive discerning of the severity level of glaucoma through clinical fundus images using hybrid PolyNet

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ABSTRACT

Purpose: Glaucoma is the major leading cause of non-curable visual defects and blindness across the world. Glaucoma is one of the eye diseases which cause an optic nerve, due to increased pressure called intraocular pressure. If the damage gets worse, it causes permanent vision loss within a few years. Timely diagnosis, early and accurate prediction are the most important factors for the prevention of visual blindness. For experts, it is uncompromising task for detecting glaucoma manually. In most of the retinal images, different investigations are carried out for identifying an automated detection tool for diagnosing glaucoma.

Methods: The main aim of this paper is to build a machine learning model for predicting glaucoma and its severity level. To predict the severity level of glaucoma, firstly, the segmentation of the blood vessel is performed and classification is accomplished using proposed U-Net model. Secondly, the segmentation of the Optic Disc (OD) and the Optic Cup (OC) is performed to determine the Cup to Disk Ratio (CDR). The proposed hybrid PolyNet classifies the severity of the glaucoma disease. The earlier detection of glaucoma disease is important for preventing the severity of the eye disease.

Results: The segmentation of blood vessels is implemented using the U-Net model, the sensitivity and the specificity are measured as 97% and 88% respectively. The accuracy and the F1 score have been measured as 96% and 98% respectively. Hence, the proposed deep learning architecture with hybrid PolyNet models is implemented for OD and OC separately to produce an appropriate result. By increasing the number of layers, more features are identified. The current work deals with the retinal fundus images and achieves the highest accuracy of 96.21% in the ACRIMA dataset with sensitivity and specificity of about 97.32% and 94.26% respectively.

Conclusion: Blood vessel segmentation is achieved with U-Net and the segmentation of the optic disc and the optic curve have been achieved and the severity level of glaucoma has been identified using hybrid PolyNet with high accuracy.

1. Introduction

Medical imaging is the process in the field of medicine for recreating images of the internal part of the body for diagnostic purposes. Computer vision techniques are designed and extended in the medical field, especially for Ophthalmology. Ophthalmology is the study of medical treatment and medical diagnosis of eye syndrome. Different eye-related diseases such as diabetic retinopathy (DR), glaucoma, cataract and etc., have seen advancements over 30 years. To capture the retinal images, a fundus camera is used and a picture captured from the fundus camera is referred as fundus image. These fundus images are very useful to diagnose most retinal-related diseases such as maculopathy, glaucoma,

diabetic retinopathy, cataract, and etc., There are different features related to the retinal and a few are fovea, optic disc, optic cup exudates and macula which varies from person to person. In particular, glaucoma is a neurodegenerative disease which harms the optic nerve due to pressure called intraocular pressure that is present in the eye. It mainly affects for adults over the age of 40. For the past 5 years, the loss of optic nerve fibre has increased from 6 % to 63 % [1,2]. Eventually, Glaucoma has been identified mainly by evaluating the optic disc and optic cup [4]. Semi-supervised GANs[5] model is proposed for segmenting the optic disc and cup. It includes a segmentation net, a generator, and a discriminator to map fundus images to segmentation maps.

For examining glaucoma detection, the eyes are dilated with drops to

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widen the pupils. This dilation allows ophthalmologists to study the optic nerve for any signs of damage and symptoms of the disease. OD is located at the back portion of the eye and it is a spherical area in which the optic nerve connects to the retina. The optic disc [6] is the bright, yellow part of the retina, which likely contains many neurons. The OD may slightly appear circular and it is called a blind spot for the fact that it doesn't give the colour – attiring photoreceptors namely rods and cones. The OC is the bright, cup-like area that is present in the centre of the OD.

Generally, the OD [7] is positioned 3 to 4 mm at the nasal part of the fovea with a dimension of 1.76 mm horizontally and by 1.92 mm vertically. Here, a central depression called the OC is positioned average dimension of 1.88 mm vertically and 1.77 mm horizontally. In consequence, for the normal eye, the size of the OC is one-third of the OD. The size of the OD increases compared to the normal size of the eye, which may lead to glaucoma. Blind patches appear in the eyesight when the optic nerve gradually deteriorates. Fluid typically accumulates in the front of the eye. By placing pressure on the eye, the secretion of additional fluid gradually harms the optic nerve. It could result in elevated intraocular pressure. As seen in Fig. 1.b, the intraocular pressure causes bulges in the optic disc and optic cup. The ratio between the size of the OD and OC should be using CDR. Based on the CDR value, the level of glaucoma can be calculated. An experiment on the same patient with different CDRs was measured horizontally and vertically. If more optic nerve fibres die, the size of the OC becomes larger related to the OD, it may lead to increase in the CDR.

The early detection of glaucoma will prevent vision loss. The main aim of this work is to identify glaucoma and classify its severity level. Pre-processing methods such as normalization modified filtering and CLAHE are processed and the features such as blood vessels, OD and OC are segmented. In retinal fundus image, the segmentation of OD and OC are performed to calculate the CDR. Glaucoma disease leads to blindness in human beings worldwide and stands third in causing blindness in India. Thus, early detection of glaucoma disease is essential for preventing eye diseases from getting worse. Glaucoma is said to increase the CDR in the eye, affecting peripheral vision. A variety of image processing techniques that are used in diagnosing glaucoma based on Cup to Disc ratio evaluation of the pre-processed retinal fundus image are addressed in the existing work. The computational algorithms are evaluated on the public fundus image datasets and the performance results are compared. The Sample of a normal fundus eye image is shown in Fig. 1(a) and an eye image affected with glaucoma is shown in Fig. 1(b).

The main contribution of this paper is to work with different datasets such as Drishti-GS1, REFUGE, ACRIMA, ORIGA, RIM-ONE and the real-time data sets. Segmentation of blood vessels is carried out with the proposed U-Net model, and segmenting the optic disc and the optic curve are accomplished with the proposed hybrid PolyNet model to predict the severity of the glaucoma. The pre-processing is accomplished on the retinal fundus image by normalization, filtering and template matching. After pre-processing, the unnecessary noise is removed and this resultant image is moved to perform the feature extraction. In

feature extraction, the optic disc is segmented to capture the optic cup accurately. Once the optic disc is extracted, the optic cup is segmented to find the CDR which identifies the severity level of glaucoma. For accuracy, deep neural network models are used to predict the classification of glaucoma or non-glaucoma and the severity level of the glaucoma. Fig. 2 represents the proposed workflow for identifying the severity level of glaucoma disease.

This paper consists of five sections. Section II describes the methods and materials and it explains the complete explanation about dataset. Section III describes the experimental setup for the current work. Section IV describes the results obtained from the proposed system and is compared with the existing system. Section V summarizes the proposed work and notifies the enhancement of the work which would be carried in the future research.

2. Methods and Materials

2.1. Collection of data

For analysing and investigating the severity level of glaucoma, a large number of datasets are collected and applied through deep learning techniques. The datasets are labelled by the ophthalmologists with two classes such as glaucoma and non-glaucoma. The deep learning approach was applied to the databases to assess the viability of the technology through the performance on images with varying resolutions. The image obtained from the class sample was classified randomly into entirely non-dependent sub-groups for training and testing purposes.

For the segmentation of blood vessels, OD and OC, images that are collected from public datasets such as Drishti-GS1, REFUGE, ACRIMA, ORIGA, RIM-ONE and the real-time data sets. In DRISHTI-GS, 101 fundus images are available where 50 images are taken for training and 51 images are taken for testing. It is a high-resolution image with a width and height of 2896×1944 , which is collected from Indian hospitals and labelled by the Indian experts. A mask of OD has been redeemed by four ophthalmologists with different years of experience. In the REFUGE dataset, 1200 fundus images are available, of which 1080 images are normal and the remaining 120 images are affected with glaucoma infections. In the ACRIMA dataset, 705 fundus images are available, of which 309 images are normal and the remaining 396 images are affected with glaucoma. In the ORIGA dataset, 650 fundus images are available, of which 482 images are normal and the remaining 168 images are affected with glaucoma disease. In RIM-ONE (r3), the dataset contains 159 fundus images with two manual segmentations of each fundus image. Here the optic disc is cropped and magnified around the OD and OC. In RIM-ONE, the dataset contains 173 fundus images, of which 118 images are normal eye, 18 images are severe, 14 images are moderate, 12 images has mild glaucoma and 11 images represents Ocular Hyper Tension (OHT) images. The real time dataset collected from Aravind Eye hospital contains 150 retinal fundus images between the age group 20 and 70 years. The collection of the dataset for the segmentation of blood vessels, extraction of the optic disc and optic

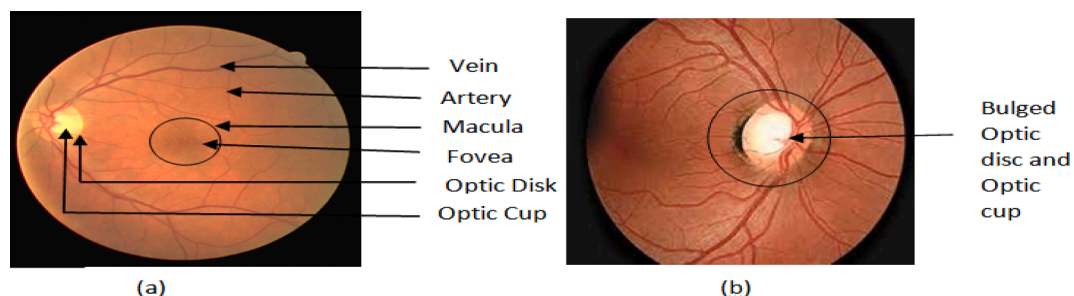


Fig. 1. (a) Normal Fundus Image (b) Abnormal affected by Glaucoma.

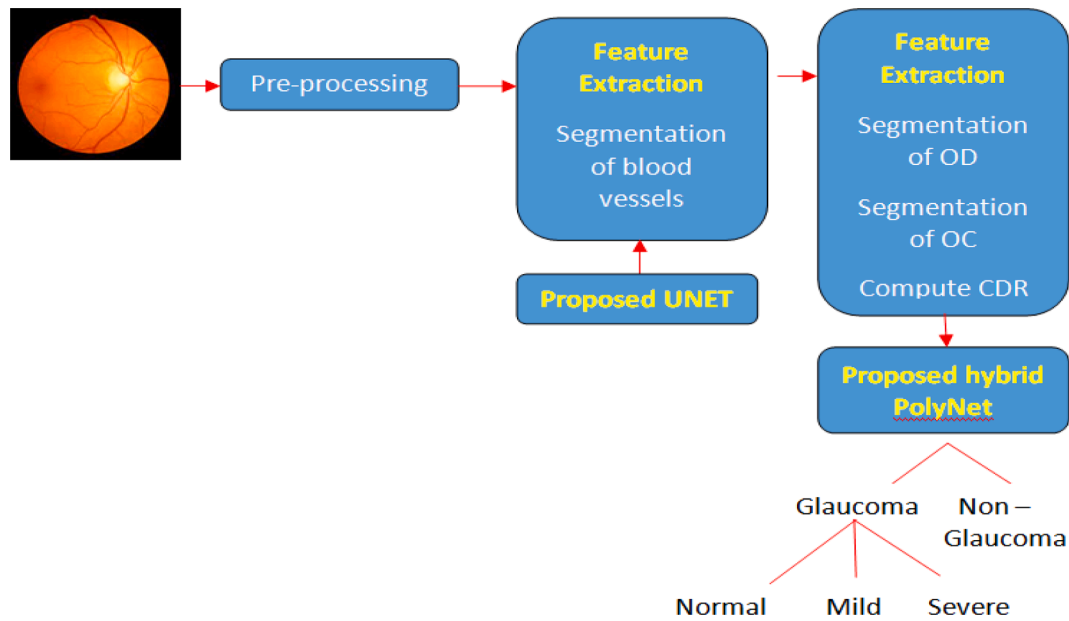


Fig. 2. Proposed workflow for identifying the severity level of the glaucoma disease.

curve and finding out the severity level of the glaucoma disease are shown in Table 1. Fig. 3 shows the graphical view of the database collection.

2.2. Pre-processing techniques

The pre-processing techniques are performed for noise removal in the image, restoration of an image, calculating morphological operations to change the original image to a binary image or grayscale image. The main goal of this technique is to enhance the quality of image data by removing unwanted falsehoods. The techniques used in the proposed preprocessing are explained below:

2.2.1. Normalization

After image data collection from Drishti-GS1, REFUGE, ACRIMA, ORIGA, RIM-ONE and the real-time data sets, a pre-processing method is performed using normalization on glaucoma image. A maximum of the retinal fundus image are captured from a fundus photography machine which are prone to two noises commonly impulse noise which is related to electrical noises and machinery noise which is related to machinery noise. This methodology is very effective in taking out almost all the electrical and machinery noises, thus helping in enhancing the image quality significantly [8]. The normalization process develops the image quality by removing the annoying noise present in the image. The formula for calculating the normalization of an image is given in equation (1).

$$ImageNormalization = (Image - Min) \frac{newMax - newMin}{Max - Min} + newMin \quad (1)$$

The image is represented as the original glaucoma image. Initially assigning $Min = 0$ and $Max = 255$ and it is indicated as the intensity range of the original fundus image and $newMax$ and $newMin$ is the intensity range of a noised removed image. The intensity range of the original image will be 50 to 180 and once the normalization process is applied, 50 will be subtracted from each pixel intensity. Later every pixel intensity is multiplied by $255/130$, to get the range from 0 to 255. The new image after normalization will be between 0 and 255.

Correspondingly, after acquiring images from Drishti-GS1, REFUGE, ACRIMA, ORIGA, RIM-ONE and the real time data sets, the normalization method regularizes and removes all the unnecessary background noises from the acquired retinal images. Certain morphological operations like contrast enhancement and binarization preserve the FOV of the images and later down-sampling process is performed. CLAHE is an effective technique for contrast enhancement by limiting the maximum slope from the function of transformation. Needless to apply the histogram equalization as it is applied to little non-overlapping regions. Henceforth, the nearby tiles are joined using bilinear interpolation to minimize the boundaries. However, overfitting due to the small number of the dataset is avoided by the data augmentation process. The data augmentation increases the variation within the training dataset. This process is repeated to crop, flip, and translate the images down to the scaled size of 227×227 .

Table 1

Dataset Collection for the segmentation of Blood vessels, OD and OC.

Dataset name	Available	Number of Images			Information
		Normal	Glaucoma	Total	
DRISHTI-GS1	Online	31	70	101	Classified according to the labels of normal and glaucomatous
REFUGE	Online	1080	120	1200	Pixel-wise observations of OD and OC, Masking of Fovea, Classified according to the labels of normal and glaucomatous
ACRIMA	Online	309	396	705	Classification labels of normal and glaucomatous
ORIGA	Online	482	168	650	Segmentation of OD and OC, Classified according to the labels of normal and glaucomatous
RIMONE	Online	92	39	131	Manual segmentation masks of OD and OC, Classified according to the labels of normal and glaucomatous
REAL- TIME DATASET	Aravind Eye Hospital, Coimbatore	46	104	150	Classified according to the labels of normal and glaucomatous
Total		2040	897	2937	Classified according to the labels of normal and glaucomatous

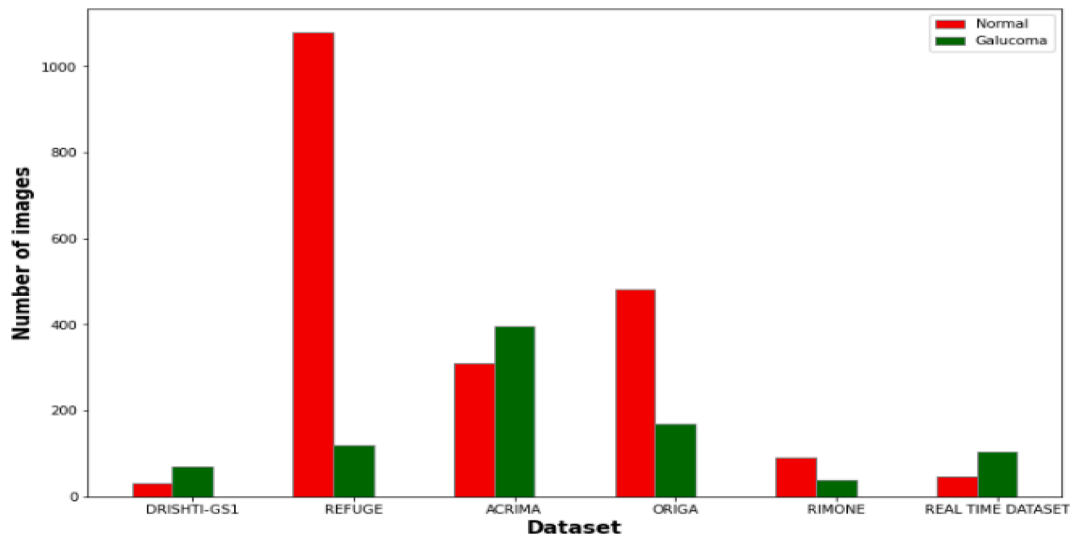


Fig. 3. Dataset for OD and OC.

2.2.2. Modified wiener filter

It is one of the restoration filters that accomplish optimal bartering between filtering and noise smoothing. It takes off the additional noise and the inputs in the blurring simultaneously. Fig. 4 shows the original image, motion blurred image, Motion blurring, and Gaussian noise where the image is applied with a wiener filter. L is a constant variable for optimizing the resultant. Equation (2) is obtained from the least square method.

$$Fest(x, y) = |A(x, y)|^2 \cdot \frac{B(x, y)}{(|A(x, y)|^2 + L(x, y))} \quad (2)$$

2.3. Blood vessel segmentation

A crucial stage in the processing of colour retinal fundus pictures used in ophthalmology to monitor and diagnose retinal diseases such as glaucoma, diabetic retinopathy and age-related macular degeneration is segmenting blood vessels. When retinal tissue has completely been lost due to very advanced glaucoma, a condition known

as bayoneting or “bean-pot” cupping may occur where retinal vessels disappear when they make a sharp turn into the cup. Deeper vessels might not be in focus as they reappear. By having routine eye exams, you can prevent or catch more serious issues early on, such as glaucoma, which can cause blood vessels to burst.

The early identification of glaucoma will prevent vision loss. The main focus is to detect glaucoma and classify the severity level. Pre-processing methods such as normalization, modified filtering and CLAHE are processed and features such as blood vessels, optic disc and optic cup are segmented. In retinal fundus image, blood vessels are segmented and the resultant output is given as an input to the segmentation of OD and OC to calculate the cup to disc ratio. The proposed work for the blood vessel segmentation is shown in Fig. 5.

The normal fundus image is rehabilitated to RGB image and the masking images are transformed into a gray image. From the RGB image [12], the grayscale is predicted by using equation (3), to reduce the processing time of the image. The conversion of RGB to grayscale is obtained by the following equation, in which the different weights of RGB components are selected for the conversion process.

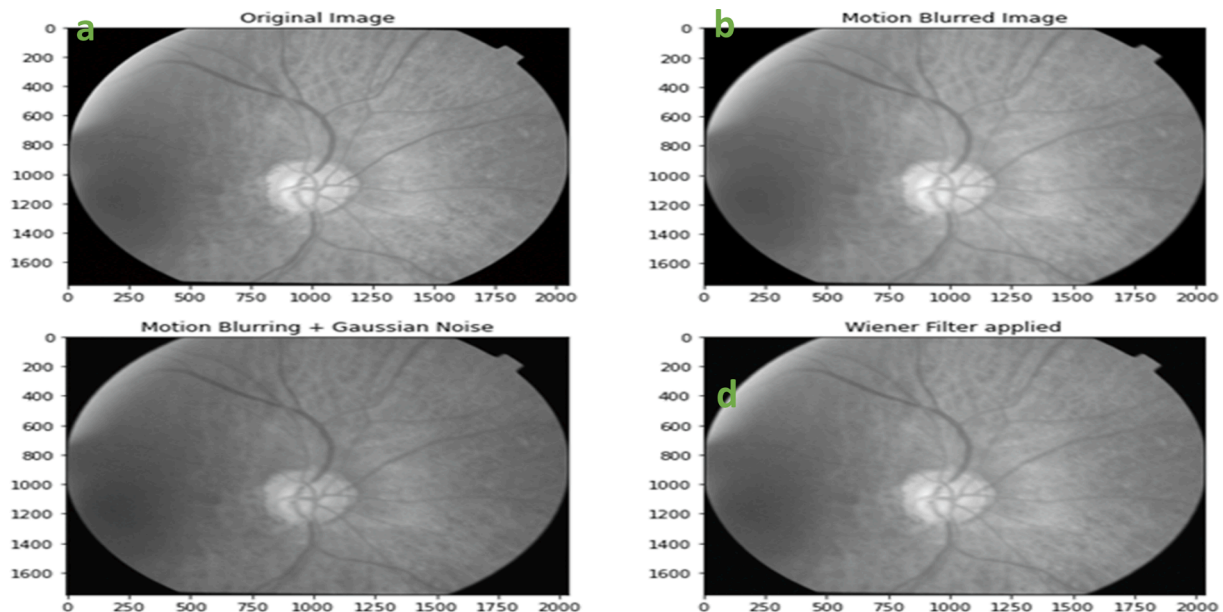


Fig. 4. (a) Original image, (b) Motion blurred image, (c) Motion blurring (d) Gaussian noise.

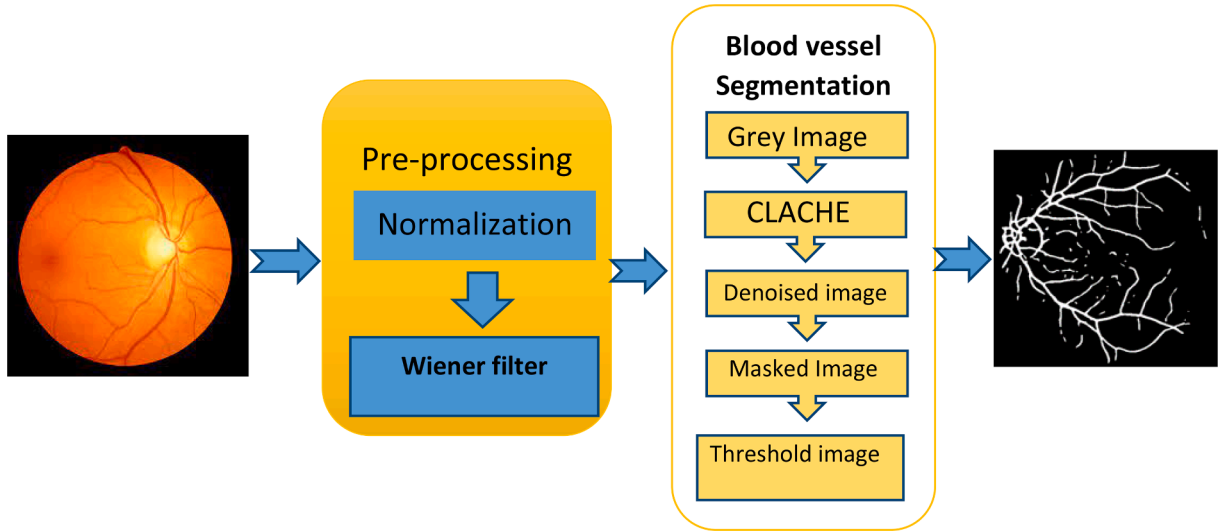


Fig. 5. Proposed workflow for Blood Vessel Segmentation.

$$\text{Gray Level} = 0.299 \cdot R + 0.587 \cdot G + 0.114 \cdot B \quad (3)$$

where R, G, and B represent the Red, Green, and Blue components of the retinal fundus image respectively. The green channel gives more information of the retinal fundus image so the weight of this channel is larger when compared to the other channels.

The averages of the RGB values are calculated. Then, the total sum of the difference between the RGB values is calculated from the sum value and the mask value using equation (4).

$$\text{mask}(a, b) = \begin{cases} 1, \text{avg}(R, G, B)T_1, K < T_2 \\ 0, \text{otherwise} \end{cases} \quad (4)$$

where $K = |R - G| + |R - B| + |G - B|$, T_1 and T_2 are the threshold values and here T_1 and T_2 value are assigned as 40 and 50 to get the best performance. Fig. 6 shows the RGB image, gray image and masking image.

CLAHE is used to enhance the contrast of the image. CLAHE works on small, small regions in the image called tiles. The neighbouring images are joined using bilinear interpolation to eliminate the boundaries. CLAHE is applied to the grey image and is named hist_eq. The histogram of the retinal fundus image has intensity levels of range from 0 to L-1

$$\text{his}(a_k) = n_k \quad (5)$$

where a_k is the value of intensity, n_k is the number of pixels with intensity a_k , hist a_k is the histogram of the retinal fundus image with grey scale level a_k .

Next, the cumulative histogram is calculated using the equation (6).

$$C(h_i) = \sum_{i=1}^k \text{his}(r_i) \quad (6)$$

where k is the greyscale image, $\text{his}(r_i)$ is the number of pixels in the i^{th} grey level, and $C(h_i)$ is the cumulative histogram value of the i^{th} grey scale.

The gamma value is taken as 1.2 to construct the lookup table by plotting the pixel values ranges from 0 to 255. After calculating the table, apply the gamma correction method with the help of the lookup table. The gamma value is calculated by equation (7).

$$g = \log \left(\frac{\text{median} - \text{minimum}}{\text{maximum} - \text{median}} \right) \quad (7)$$

Then the g value is normalized from 0.8 to 1.2 and it is given in the

$$\text{equation.} \begin{cases} 0.8, & \text{if } g < 0.8 \\ 1.2, & \text{if } g > 1.2 \\ g = & \text{otherwise} \end{cases}$$

Here, the gamma value is limited to 1.2. By the normalized gamma value, the contrast of the grayscale image is enhanced with the gamma correction method. The contrast enhancement of the image is computed by equation (8).

$$G(a, b) = (f(a, b))^{\frac{1}{g}} \quad (8)$$

where f (a, b) is the intensity of the input image.

For denoising, fast non-local means denoising is applied to the

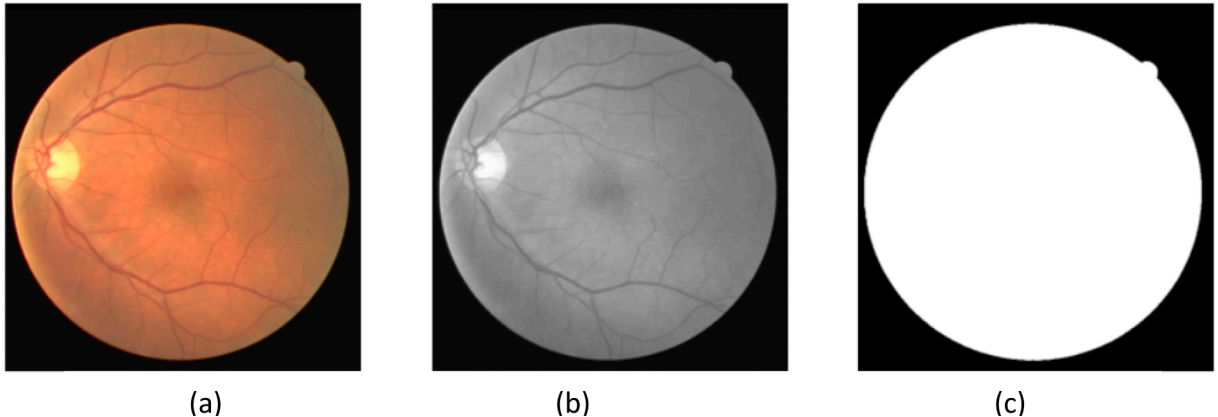


Fig. 6. (a) RGB image, (b) Gray image, (c)Masking noise.

gamma-corrected grayscale image. Assume Ω is the specific part of the image, x and y are the two points present inside the image, then the equation (9) is

$$u(x) = \frac{1}{C(x)} \int_{\Omega} v(y) f(x, y) d_y \quad (9)$$

where $u(x)$ is the filtered value at point x in the image, and $v(y)$ is the unfiltered value at point y in the image. $f(x, y)$ is the weighting function and the integral part is evaluated as $\forall y \in \Omega$.

$c(x)$ is the factor of normalisation and it is given by the equation (10)

$$C(x) = \int_{\Omega} f(x, y) d_y \quad (10)$$

Fig. 7 shows the h_{is_eq} , gamma-corrected and denoised images.

The ridges are detected using the Hessian matrix and its eigenvalues. Here, the sigma value is taken as 3.0 to get the better boundary edge detection. Hessian matrix returns the positive values for intensity whereas negative values are for ridges and small values having small noise regions as shown in equation (11). The eigenvalues are computed with the equation (12). Here, the membrane boundaries, ridges and blobs are to be detected with the equation (13).

$$H = \begin{bmatrix} L_{xx} & L_{xy} \\ L_{xy} & L_{yy} \end{bmatrix} \quad (11)$$

$$\lambda_{1,2} = \frac{1}{2} (L_{xx} + L_{yy} \pm \sqrt{(L_{xx} - L_{yy})^2 + (2L_{xy})^2}) (|\lambda_1| \leq |\lambda_2|), \quad (12)$$

$$\Delta(L) = \text{trace}(H) = L_{xx} + L_{yy} = \lambda_1 + \lambda_2 \quad (13)$$

Fig. 8 shows the denoised images and its corresponding grey and their sigma image.

The denoised images are then applied for the normalization stuff, to overcome the negative value. With the resultant image, masking is applied. Next the cover ratio has been identified to detect the exact part of the white pixels were that are covered. The threshold value is set as 165 and the operation performed is THRESH_BINARY_INV. Fig. 9 shows the masked images, median and thresholded image.

The accuracy of the threshold and labelled image is predicted as 0.9597. Median image is in the form of grey such that the median image is in the form array. Further, the `cmap` function rules the colormap with the grey and displays the values.

Fig. 10 shows the threshold image, the labelled image and the median image.

Fig. 11 shows the RGB, threshold and labelled images of the different dataset.

2.4. Segmentation of optic disc and optic cup

Glaucoma is a disease, it affects the optic nerve and is caused by the intraocular in the eye. It mostly affects the person having diabetes and

age above 60 [27]. The early signs indicate vision loss, eye redness, eye pain, severe headaches, and etc. The OD is the brightest spherical like shape, present inside the retina, and it is yellow. The shape of OD is round and the shape is different from individuals [29]. The ONH connects with more neurons and is called the “blind spot” because the rods and cones in the color photoreceptors are not present [28]. In glaucoma prediction, the optic nerve head is examined by measuring CDR. For estimation, the segmentation of OD and OC is needed and it can be extracted by using computer vision algorithms [15]. For medical professionals or ophthalmologists, the extraction of OD and OC [13] is a very difficult and time consuming process for determining the CDR.

The main focus is to determine the CDR ratio and determine whether the fundus image is affected by glaucoma or not. If affected by glaucoma, it will produce the severity level of the disease. The pre-processing fundus image proceeds further for the edge detection to acquire the features of the pre-processed image. Here, the canny edge detection method and the watershed algorithm are used to confine the specific part of the optic nerve. In the located region, if glaucoma is identified, then the affected part alone is cropped by 512×512 pixels [3]. For extracting the regions of OD and OC, two separate and convolutional neural networks are used with 16 layers. The cropped image is set as the input for both the CNN model and will generate a masked image of OD and OC. With this predicted mask image CDR is calculated to predict the severity level of glaucoma. Fig. 12. shows the proposed workflow for the segmentation of OD and OC.

2.4.1. Extraction of shape

Shape detection is mainly performed by Edge Histogram Descriptor (EHD) [26]. The area of the region is captured with different incidences of an image. The input image is split into 4×4 segments and is specified as sub images and these images are again separated into 16 blocks [30] which will produce the histogram for every block. In every part, the edges are available, and these edges are classified according to horizontal, vertical, 45-degree diagonal, 135-degree diagonal, and non-directional edges.

2.4.2. Feature extraction

The contour shape is segmented by the Canny edge detection method and the edge features are identified through the watershed algorithm. By extracting its features, it will help to localize the OD and OC easily. The localized part is cropped and set as an input for segmenting the OD and OC.

2.4.3. Canny edge detection

This technique is used to determine the boundary of the image. In the proposed work, this method is applied to predict and detect the proper localization of the boundaries of OD and OC. The data of the edge boundaries are extracted by using the canny edge detection method. The pixels present on the edges are lighter to brighter by the data vector. Thus, the OD and OC are easily segmented.

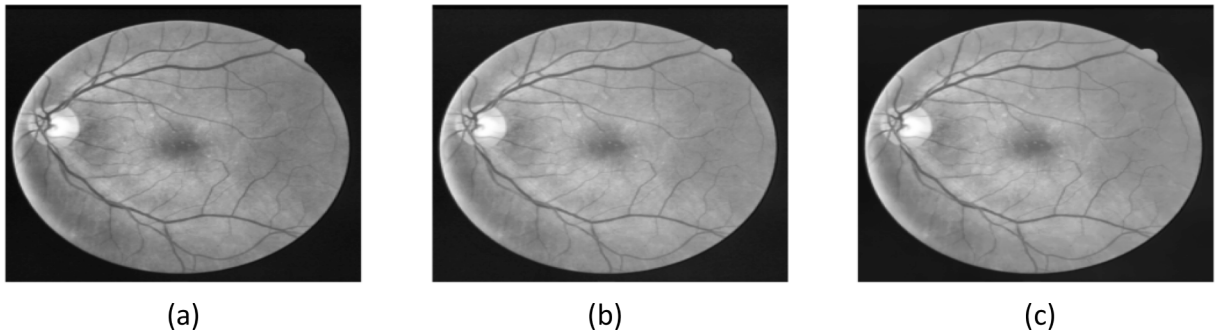


Fig. 7. (a) h_{is_eq} (b) Gamma corrected and (c) denoised images.

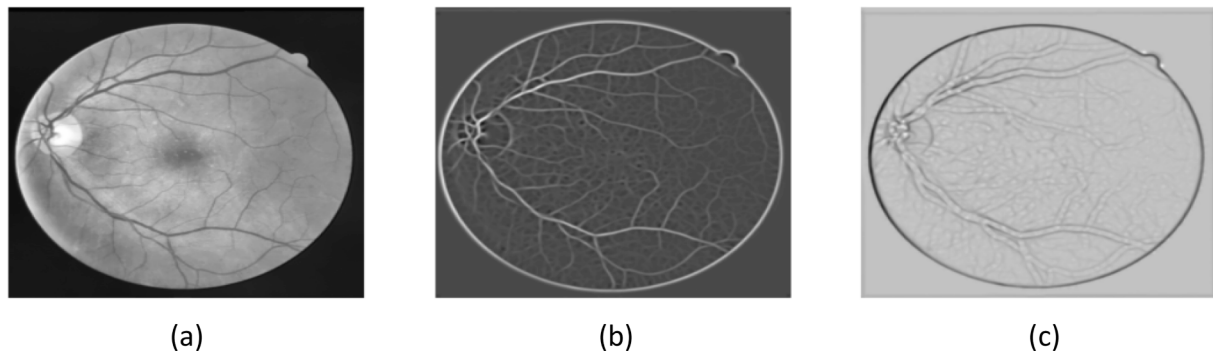


Fig. 8. (a) Denoised images (b) grey image and (c) sigma image.

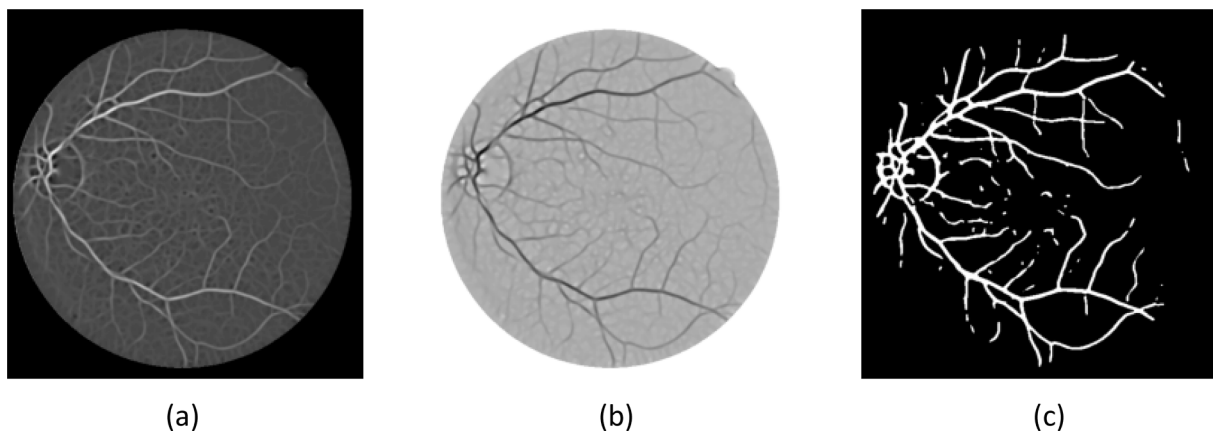


Fig. 9. (a)Masked image, (b)Median image and (c)Threshold image.

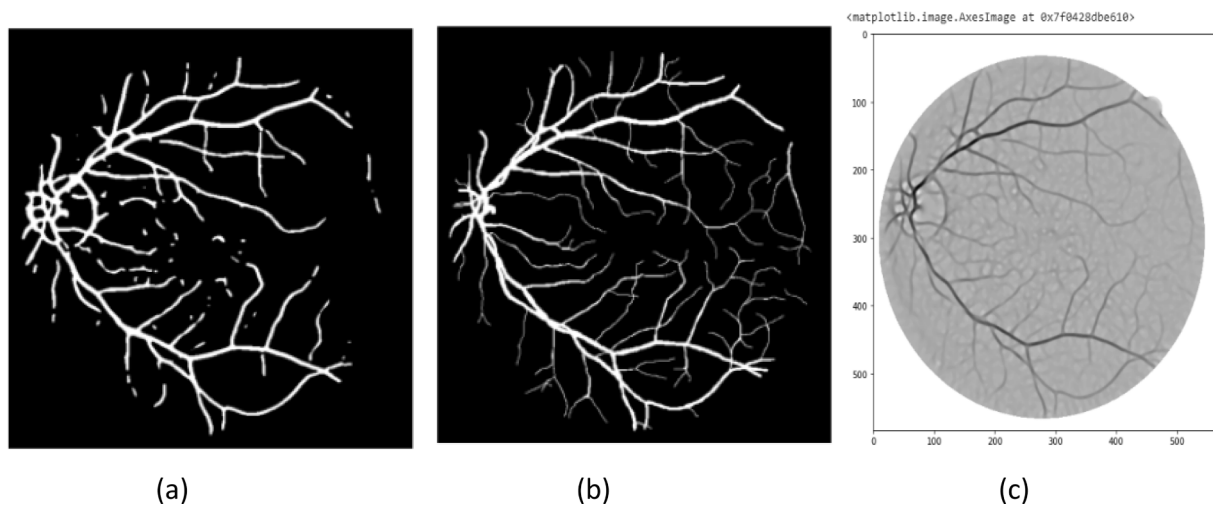


Fig. 10. Threshold image, labelled image, median image.

2.4.4. Watershed algorithm

In the proposed work, contour shape detection is performed by using the watershed algorithm. Based on this outcome, the boundaries and the lines of OD and OC are extracted. The main task of the watershed algorithm is the input image which is split into different sets according to the shapes of the contour. The pixels of each part are grouped and are named as superpixels. In the entire surface of the image, watershed algorithm predicts the watershed lines and the catchment basins [22,23]. There are different algorithms to identify the region by observing the dark and light pixels that are low and high respectively. With this

information, the image is cropped and this cropped image is given to the algorithm for edge and boundary extraction of OD and OC [25]. Cup Disc Encoder Decoder Network (CDED-Net) [24] for joint Optic Disc (OD) and Optic Cup segmentation (OC) are performed. The pre-processing and post-processing steps are eliminated to lower the system cost.

Hence the image is cropped into 512×512 , which leads to data loss. But the remaining part is regardless for the prediction of glaucoma, so it may lead to reduced computational time and an increase in accuracy for segmenting the OD and OC. Fig. 13 shows the pre-processed image after

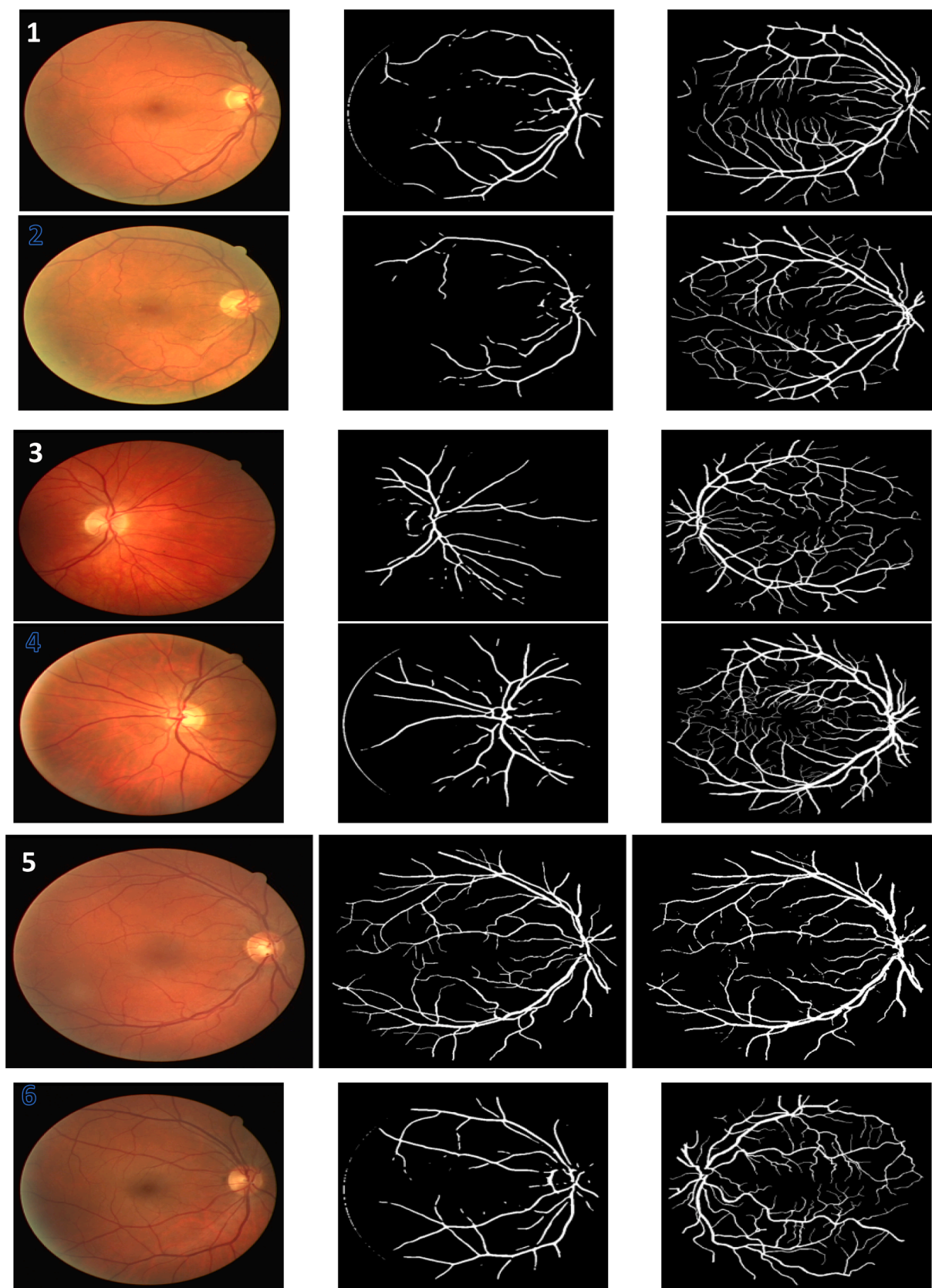


Fig. 11. RGB, threshold and the labelled images of the different datasets.

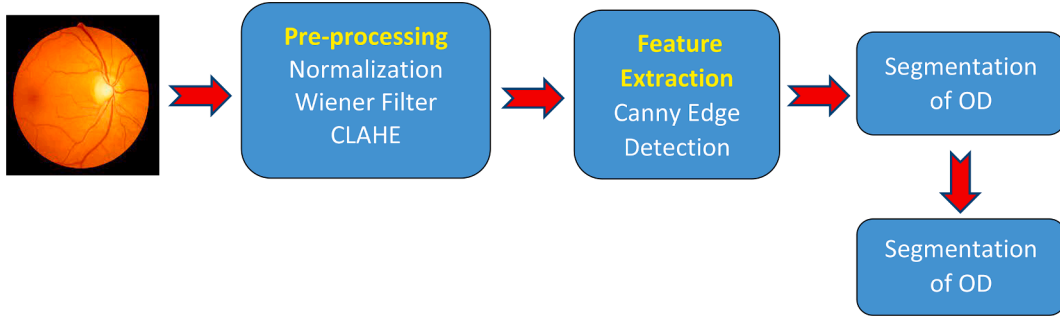


Fig. 12. Proposed workflow of the segmentation of OD and OC.

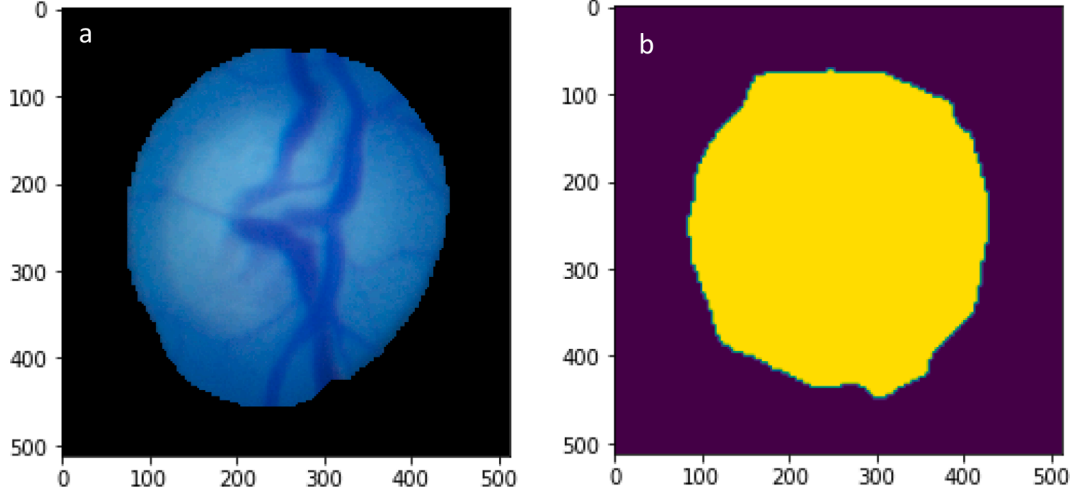


Fig. 13. (a) Pre-processed image after applying CLACHE (b) OD detection after applying the watershed algorithm.

applying CLAHE and the OD detection after applying the watershed algorithm.

2.5. Cup-to-Disc Ratio(CDR)

The Cup-to-Disc Ratio (CDR) in Eqn 14 is determined by dividing the maximum diameter of the optic cup by the maximum diameter of the optic disc and is calculated after the optic disc and optic cup have been segmented.

$$\text{Cup-to-disc ratio(CDR)} = \frac{\text{Diameter of cup}}{\text{Diameter of disc}} \quad (14)$$

The estimation of the mean error of CDR and the standard deviation of CDR estimation errors are analysed with the correlation coefficient ω which measures the linear correlation between the estimated CDRs and the manual CDRs. Here, the correlation coefficient ω can be calculated using equation (15),

$$\omega = \frac{\sum_{i=1}^N (CDR_i^{Es} - \overline{CDR^{Es}})(CDR_i^{Ma} - \overline{CDR^{Ma}})}{\sqrt{\sum_{i=1}^N (CDR_i^{Es} - \overline{CDR^{Es}})^2 \sum_{i=1}^N (CDR_i^{Ma} - \overline{CDR^{Ma}})^2}} \quad (15)$$

CDR_i^{Es} and CDR_i^{Ma} are the CDR values from the estimated and manual respectively, N is the number of each observed data, $\overline{CDR^{Es}}$ and $\overline{CDR^{Ma}}$ are the means of the CDR values.

CDR error represents the absolute deviation which measures the difference between the estimated and manual CDRs.

$$\rho_i = |CDR_i^{Es} - CDR_i^{Ma}| \quad (16)$$

Where CDR_i^{Es} denotes the i th estimated CDR and CDR_i^{Ma} denotes the

i th manual CDR. The mean absolute error is determined by the equation (17) and the equation (18),

$$\text{Mean absolute Error} = \frac{1}{N} \sum_{i=1}^N |CDR_i^{Es} - CDR_i^{Ma}| \quad (17)$$

$$\text{Mean absolute Error} = \frac{1}{N} \sum_{i=1}^N \rho_i \quad (18)$$

The stages of glaucoma, such as mild, moderate, and severe were predicted using the images in the databases. Fig. 14 and Fig. 15 demonstrate the segmentation of the OD and OC from the retinal fundus image. If the CDR value is higher than 0.5, the diagnosis of glaucoma is made otherwise, the CDR value lower than 0.5 indicates that the eye is healthy.

3. Experimental Setup

3.1. Proposed U-Net

The segmentation of medical images frequently employs U-Net. Skip connections are used in the U-shaped architecture to merge the low-level detailed feature maps from the encoder and the equivalent high-level semantic feature maps from the decoder. For classifying the normal and the severity of glaucoma, the proposed U-Net classifier is used. It allows image classification and identification to arrange the pixels related to the objects of its exact shape. U-Net [11] is best for predicting the severity of the glaucoma keenly as it can distinguish the normal, mild and severe stages of glaucoma perfectly. It is shaped “U” and it consists of contracting path and expansive path. An expansive path is

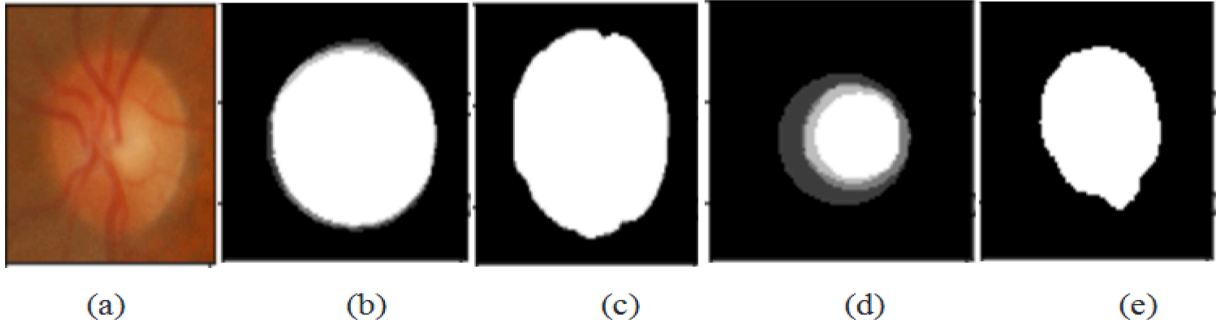


Fig. 14. (a) Original image, (b) Disc original, (c) Disc, (d) Cup original and (e) Cup.

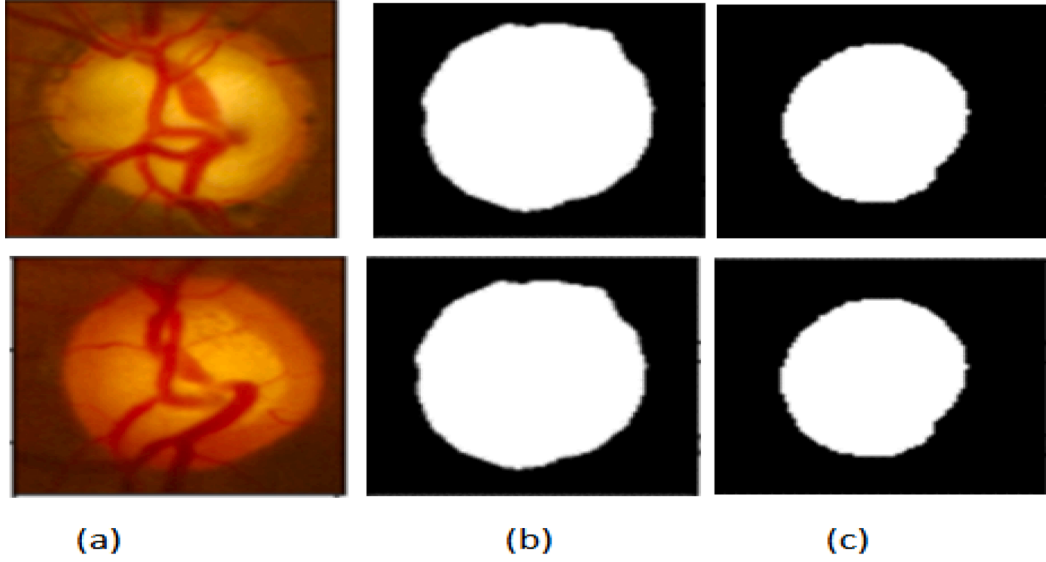


Fig. 15. (a) cropped image (b) after OD segmentation (c) after OC segmentation.

used for the up sampling process. CNN framework is used to support the extraction and evaluation of the suspicious skin region by using VGG-UNet[37].

In the proposed work, the activation layer is chosen as MISH which is a Self Regularized Non-Monotonic Activation Function. The smoothness of MISH which aids in efficient optimization and generalisation, also contributes significantly to the explanation of the improvement in results. This function helps to prevent the growth in the computation and it does not activate all the neurons at the same time. The equation for the MISH function is specified in (19),

$$f(x) = x \cdot \tanh(\ln(1 + e^x)) \quad (19)$$

The first order derivative of the MISH function is specified in equation (20),

$$\frac{d}{dx}(x \tanh(\ln(1 + e^x))) = \tanh(\ln(1 + e^x)) + \frac{e^x \operatorname{sech}^2(\ln(1 + e^x))}{1 + e^x} \quad (20)$$

The following are the steps for finding out the first order derivative of the MISH activation function

$$\frac{d}{dx}(x \tanh(\ln(1 + e^x))) \quad (21)$$

The product rules are applied for the equation (21), and the product rule is mentioned in (22)

$$(f \cdot g)' = f' \cdot g + f \cdot g' \quad (22)$$

$$f = x, g = \tanh(\ln(1 + e^x)) \quad (23)$$

$$g = \frac{d}{dx}(x) \tanh(\ln(1 + e^x)) + \frac{d}{dx} \tanh(\ln(1 + e^x)) x \quad (24)$$

$$\text{We know that } \frac{d}{dx}(x) = 1$$

$$\frac{d}{dx}(\tanh(\ln(1 + e^x))) = \frac{e^x \operatorname{sech}^2(\ln(1 + e^x))}{1 + e^x} \quad (25)$$

$$1 \cdot \tanh(\ln(1 + e^x)) + \frac{e^x \operatorname{sech}^2(\ln(1 + e^x))}{1 + e^x} x \quad (26)$$

The first order derivative is calculated in equation (26) and it is simplified, shown in equation (27),

$$1 \cdot \tanh(\ln(1 + e^x)) + \frac{e^x \operatorname{sech}^2(\ln(1 + e^x))}{1 + e^x} x : \tanh(\ln(1 + e^x)) \\ = \frac{e^x \operatorname{sech}^2(\ln(1 + e^x))}{1 + e^x}$$

$$\frac{d}{dx}(\tanh(\ln(1 + e^x))) = \tanh(\ln(1 + e^x)) \frac{e^x \operatorname{sech}^2(\ln(1 + e^x))}{1 + e^x} \quad (27)$$

Fig. 16 shows the graph of the first order derivative of the MISH activation function.

In U-Net, the input is given as any scalar value with 3 dimensional input vectors. Dropout is used to prevent our proposed model from overfitting [17,18]. It randomly sets the outgoing edges of hidden units to 0 at every execution of the training phase. In the proposed model, the layers such as convolution (19), max pooling (5), upsampling (4),

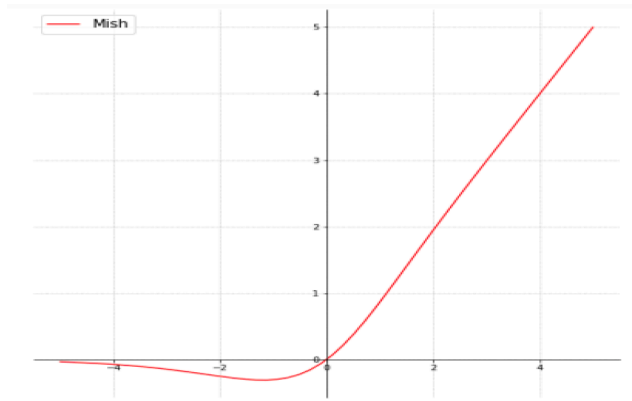


Fig. 16. Graph of the first order derivative of the MISH activation function.

dropout (18) and merge layers (4) and output with softmax activation (1). Finally, the activation function is chosen as sigmoid for convolution of the layer conv9. To overcome the image resolution problem, for every downsampling, upsampling is added. This upsampling helps to recover lost information. The model is compiled by the nadam optimizer and is fixed with a learning rate of 0.01. Here, 10-fold cross-validation is chosen to get the accurate performance of the data. The complexity of the proposed UNet model is $O(n)$. Fig. 17 shows the architecture of the proposed U-Net.

3.2. Proposed hybrid PolyNet

In the proposed work, the proposed hybrid PolyNet is applied for analysing the severity of glaucoma. The hybrid PolyNet processes are used to improve the accuracy of finding the severity of glaucoma. The cropped ONH part of size 128×128 pixels is given to the proposed model to decrease the time of execution. In the proposed model, the layers such as convolution, batch normalization, max pooling, fully connected, and output with softmax activation. The different filter sizes are used in different layers such as 64, 128, 256 and 512. The convolutional layer is to convolute two sets of data and generate the feature map with different filters. The dropout layer is used after every convolutional to avoid overfitting [21]. The Conv2D layer is responsible for the generation of a convolution kernel, which, when combined with the layer input, results in a tensor of outputs.

Pooling layers are used to minimise the dimensionality of the feature maps. The Max Pooling process is a sort of convolution in which the Kernel seeks to discover the region of the image that contains the highest

possible value. Batch normalization is responsible for taking the outputs of the previous hidden layer and standardising them before sending the input of the hidden layer that comes after it. MISH activation function is to train the model efficiently and to provide better performance. The nadam optimizer stands for Nesterov-accelerated Adaptive Moment, which is an extension of the Adam optimizer. The nadam optimizer is used to improve the overall performance of the optimization procedure. The learning rate is fixed at 0.01 and the batch size is as normally as 16. The 500 number epochs is given in the training phase.

Fig. 18 shows the structure of the proposed polyNet.

In the first inception layer, the Convolution layer (12), max pooling layer(2), concatenate(2), and Dense (1) are used. The output of the inception layer is passed into the max pooling, and the output of the max pooling is again passed to the inception layer to extract more features. The output of the second inception layer is later passed to the average pooling, which is further flattened, dropout and dense to get the desired output effectively.

MISH activation function is to train the model efficiently and to provide better performance. Here, MISH activation is used for the dense layers of 4096 units so that it stops to forwarding the negative values. In a 3-layered network, MISH is shown to consistently improve the varying dropout rates from 0.2 to 0.75 for a single dropout layer. After performing the flattened process, a 3-unit dense layer is used at the end with activation to predict the three classes of glaucoma such as mild, moderate and severe levels of glaucoma. Up to 20 epochs, the model accuracy has been identified as 0.96 and the loss has been identified as 0.23. Up to 50 epochs, the model accuracy is identified as 0.97 and the loss is identified as 0.10. For glaucoma classification, the proposed hybrid PolyNet architecture is designed and it is shown in Fig. 19

3.3. Designing and training

The proposed work uses hybrid PolyNet models for classifying the severity level of glaucoma. Significantly, it is used to increase the accuracy of classifying the severity of the disease. The simulation is done on a powerful computer that has a Ryzen 5 4500 CPU, 16 GB of RAM, and a 4 GB GPU. For developing the UNet and hybrid PolyNet model, the libraries Keras and Tensorflow are used. After training, the total parameters identified are 205,179,670, trainable parameters are 205,164,581 and not trainable parameters are 15,090.

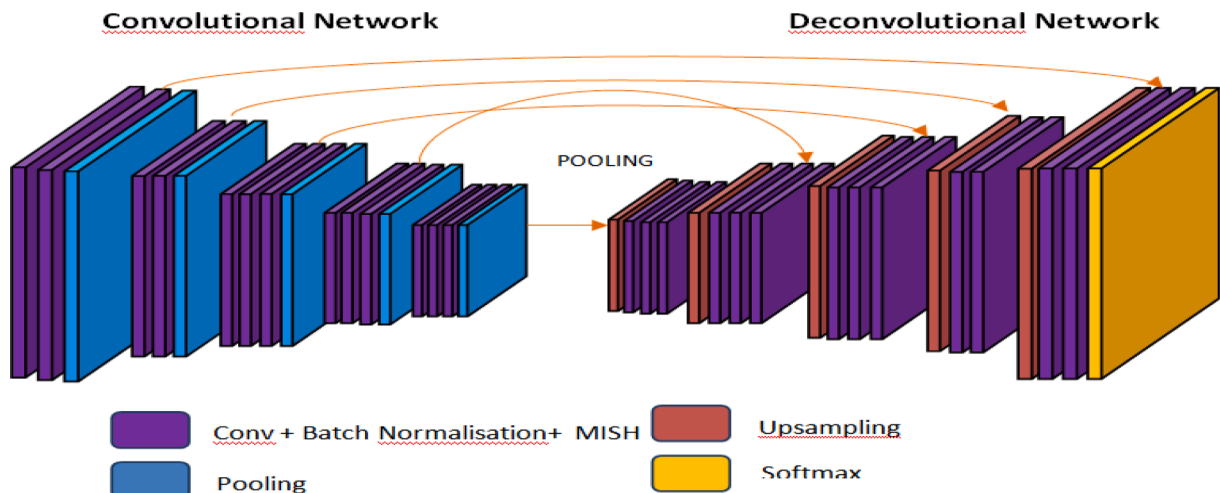


Fig. 17. Architecture of the proposed U-Net.

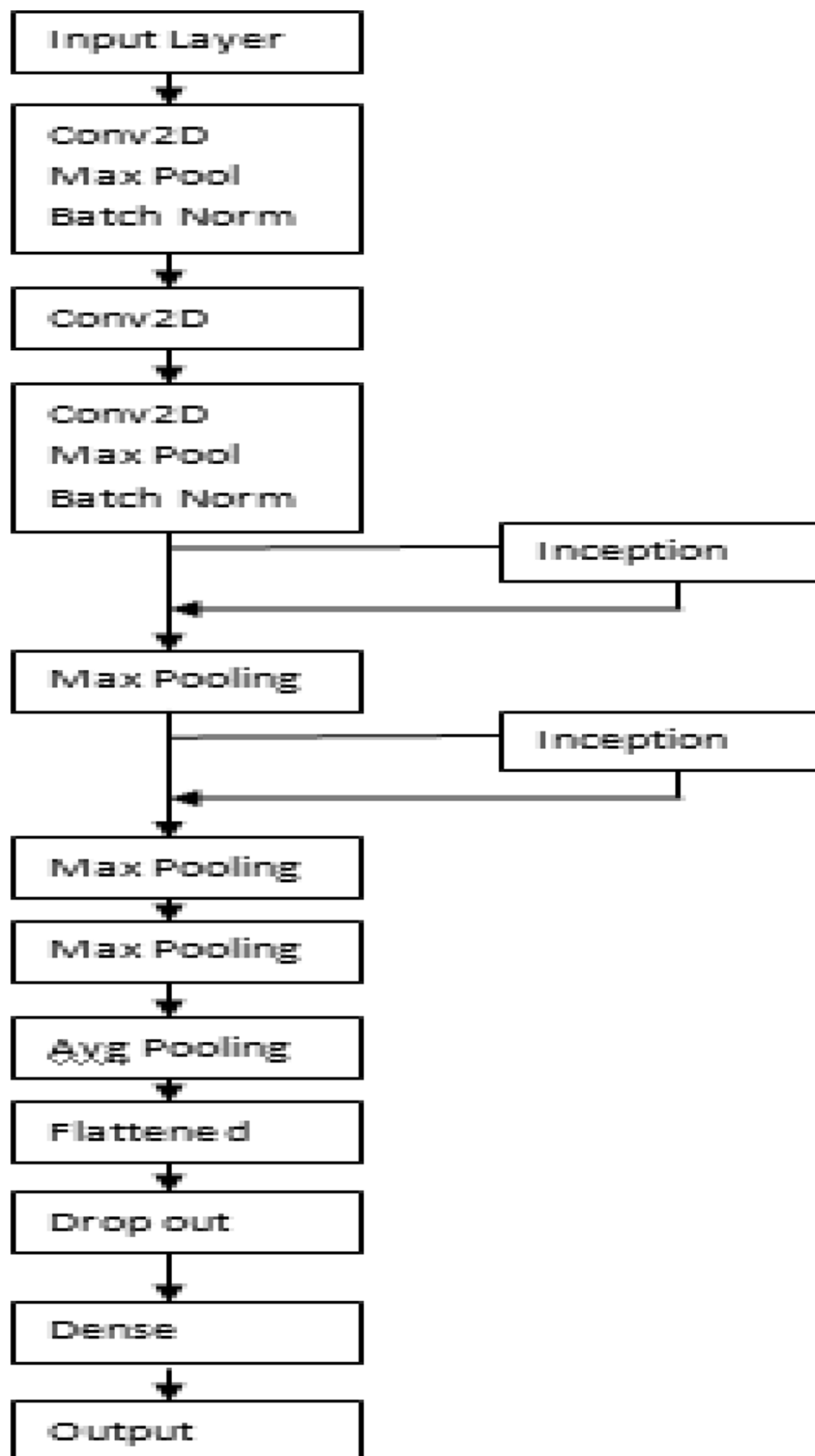


Fig. 18. Semantic structure of the proposed PolyNet.

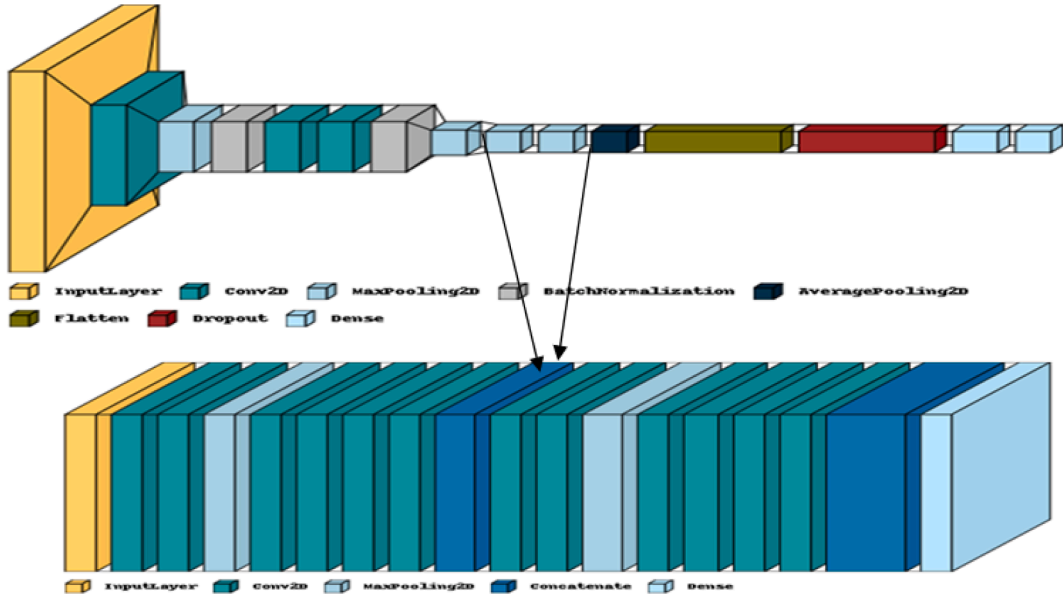


Fig. 19. Architecture of the proposed hybrid PolyNet.

4. Results and Discussions

4.1. Metric calculation for segmentation of blood vessels

The segmentation of blood vessels uses several metrics like sensitivity, specificity, precision, accuracy and F1 score are calculated. For the segmentation of the blood vessels, the results obtained from the proposed system are compared with the related work. For the blood vessel segmentation, sensitivity is used to compute the number of actual positives which are correctly predicted and specificity measure is used to compute the number of actual negatives which are predicted correctly. The specific measures for the blood vessel segmentation are computed using the Eqn 28, 29, 30, 31, 32 and 33.

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (28)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (29)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (30)$$

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \quad (31)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (32)$$

$$\text{F1 score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (33)$$

For the ORIGA dataset, the sensitivity and the specificity are measured as 89 % and 85 % respectively. Compared to the other datasets such as Drishti-GS1, REFUGE, ACRIMA, RIM-ONE and the real-time data sets, the ORIGA dataset attains the highest accuracy. Table 2 and Fig. 20 show the detailed analysis of the metrics with a different dataset. Fig. 21 shows the model accuracy and model loss for the blood vessel segmentation.

The proposed work of the blood vessel segmentation is compared with the existing work and is tabulated in Table 3.

Table 2

Metrics for the blood vessel segmentation.

Database	Accuracy	Sensitivity	Specificity
DRISHTI-GS1	74 %	76.50 %	72.50 %
REFUGE	74.67 %	75.50 %	72.60 %
ACRIMA	85.21 %	87.84 %	82.50 %
ORIGA	87.95 %	89.12 %	85.62 %
RIMONE	72.70 %	73.69 %	70.32 %
REAL TIME DATASET	81 %	82.36 %	79.56 %

4.2. Metric calculation for the segmentation of OD and OC

The segmentation of OD and OC uses several metrics such as dice similarity coefficient, Intersection over the union, Structural similarity (SSIM), sensitivity, specificity, precision, Accuracy, and F1 score has been calculated. The results obtained from the proposed system are compared with the related work.

Dice similarity Coefficient:

Considering the set P and Q, the equation for calculating the Dice metric is represented as in Eqn 34

$$DSC = \frac{2|P \cap Q|}{|P| + |Q|} \quad (34)$$

where, |P| and |Q| represent the cardinality of P and Q. Dice similarity metrics can be obtained TP, TN and FP also by Eqn 35

$$DSC = \frac{2TP}{2TP + FP + FN} \quad (35)$$

2 Intersection over Union (IOC):

It is also represented as Jaccard Index, and it is defined as the ratio of overlapping the sets of union, to measure the similarity index between the set P and the set Q, specified in Eqn 36.

$$IOU(P, Q) = \frac{|P \cap Q|}{|P| + |Q| - |P \cap Q|} \quad (36)$$

3 F1 Score:

The F1 score is represented by using Precision and Recall, it is used to measure the accuracy and also it will be used to analyse for binary classification in Eqn 37.

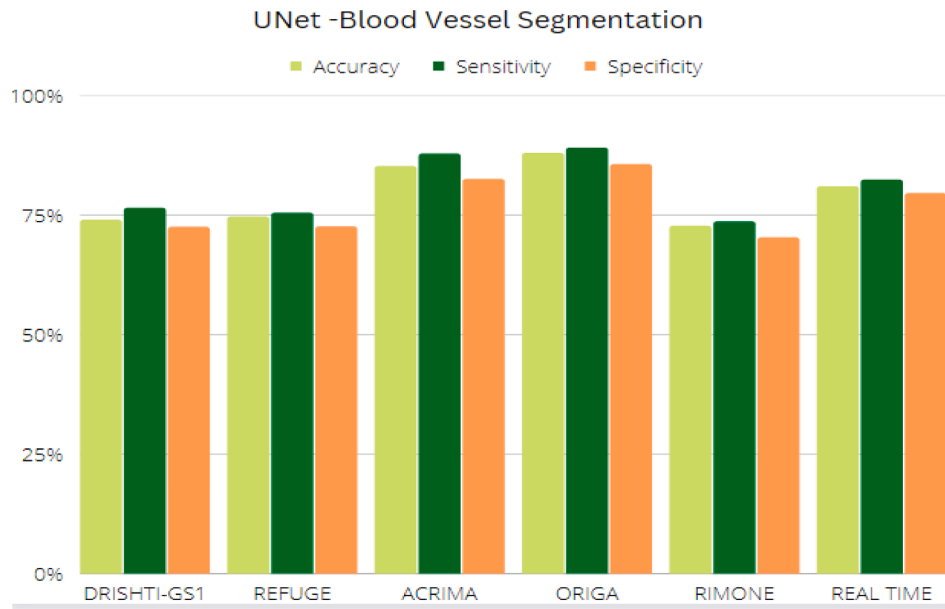


Fig. 20. Drishti-GS1, REFUGE, ACRIMA, ORIGA, RIM-ONE and the real time datasets SCORE for blood vessel segmentation.

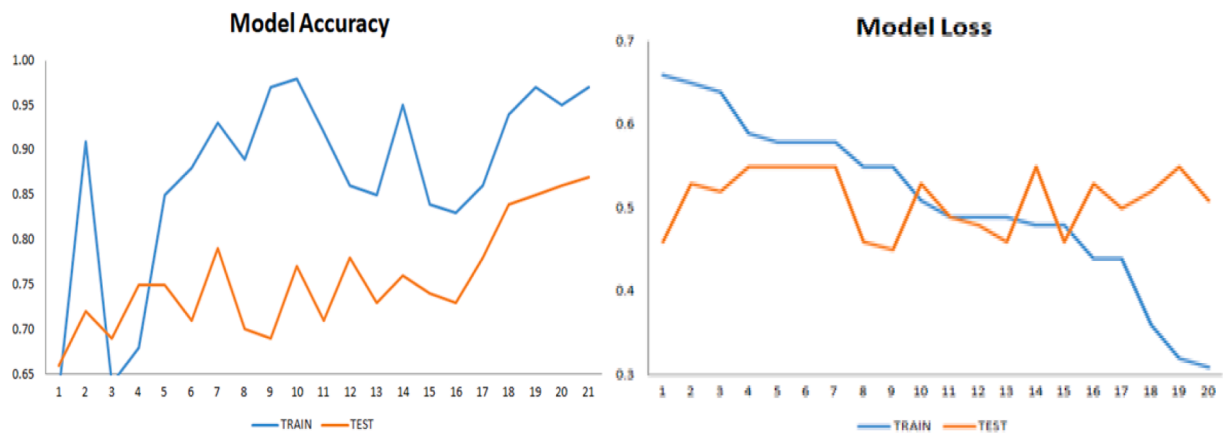


Fig. 21. Model accuracy and model loss for the blood vessel segmentation.

Table 3

Blood Vessel Segmentation-Comparison table with existing and proposed work.

REFERENCES	DATABASE	SENSITIVITY	SPECIFICITY	ACCURACY
O. Sule [9]	DRIVE	70 %	98.2 %	94.47 %
Y. Zhang and A. C. Chung [19]	DRIVE	87.23 %	96.18 %	95.04 %
	STARE	76.76 %	99.01 %	97.12 %
P. Xiuqin[10]	DRIVE	93.1 %	98.65 %	96.5 %
T. A. Soomro [16]	DRIVE	80.2 %	97.4 %	95.9 %
	STARE	80.1 %	96.9 %	96.1 %
Sangeethaa SN [14]	DRIVE	98 %	94.0	96.4
	DIARETDBO			
	DIARETDB1_V			
Proposed work	DRISHTI-GS1	76.50 %	72.50 %	74 %
	REFUGE	75.50 %	72.60 %	74.67 %
	ACRIMA	87.84 %	82.50 %	85.21 %
	ORIGA	89.12 %	85.62 %	87.95 %
	RIMONE	73.69 %	70.32 %	72.70 %
	REAL TIME DATASET	82.36 %	79.56 %	81 %

$$F1 = 2 \cdot \frac{P.R}{P + R} \quad (37)$$

4 Structural Similarity Index(SSIM):

This metric is used to identify the image quality and it is used to calculate by using the Eqn 38

$$SSIM(a, b) = \frac{(2\mu_a\mu_b + D_1)(\sigma_{ab} + D_1)}{(\mu_a^2 + \mu_b^2 + D_1)(\sigma_a^2 + \sigma_b^2 + D_2)} \quad (38)$$

where μ_a is the average value of a, μ_b is the average value of b, σ_a^2 is the variance of a, σ_b^2 is the variance of b, σ_{ab} is the covariance of a and b, and D1 and D2 are the variables.

5 Accuracy:

The numbers of pixels in the images are correctly classified in percentage by the Eqn 39

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (39)$$

All the specific measures were calculated for the segmentation of OD and OC and are displayed in Table 4.

The proposed work uses the datasets with every 500 epochs. Fig. 22 shows the dice metric, dice metric with its loss for OD segmentation of trained data for 500 epochs respectively. Fig. 23 shows the dice metric,

Table 4
Metric for OD and OC.

Metric	Disc Segmentation	Cup Segmentation
Dice Metric	0.9902	0.9835
IOU	0.9222	0.9109
F1 Score	0.9441	0.9337
SSIM	0.9472	0.9440
Accuracy	0.9882	0.9795

and its loss for OC segmentation of trained data for 500 epochs respectively. The results of 99.02 % and 98.35 % are obtained for the dice metric of the trained data for the segmentation of OD and OC respectively.

Fig. 24 shows the performance metric for OD and OC of test data for 500 epochs respectively.

Here the dice metric is common with the existing system and it is compared in Table 5. The number of layers was increased in the proposed model so that the losses were decreased.

4.3. Proposed glaucoma classification

By using the CDR ratio, the input image will be determined whether it is affected by glaucoma or not glaucoma.

If the input image is affected by glaucoma, then it is classified to find out the severity level of the glaucoma disease. The proposed model uses multiple datasets comprised of 2937 retinal images. About 70 % of the images from the dataset were augmented and obtained 205 images that are used for training. Twenty percent images from the dataset were used for validation. We trained our model in KERAS with 500 EPOCHS. For every dataset, we have calculated the confusion matrix. Compared to all the datasets, the ACRIMA dataset obtains the highest accuracy. So, here we have tabulated the Training Loss, Training Accuracy, Validation loss and Validation Accuracy only for the ACRIMA dataset.

From Table 6, it is evident only for the ACRIMA dataset, when the epoch value rises from 1 to 500, the training loss reduces from 2.0726 to 0.0740 and validation loss recedes from 2.1421 to 0.0982. Also, the training accuracy and validation accuracy rises from 0.4900 to 0.9621, and the validation accuracy also rises from 0.2421 to 0.9621. Table 7.

In ACRIMA dataset, total of 396 images are glaucomatous images. From these 331 images are mildly affected by glaucoma, 41 images are moderately affected by glaucomatous and 24 images are severely affected by glaucomatous images. The proposed hybrid polyNet classifier identified 328 mild glaucomatous images correctly and misinterpreted 3 images as moderate. It has also identified 33 moderate

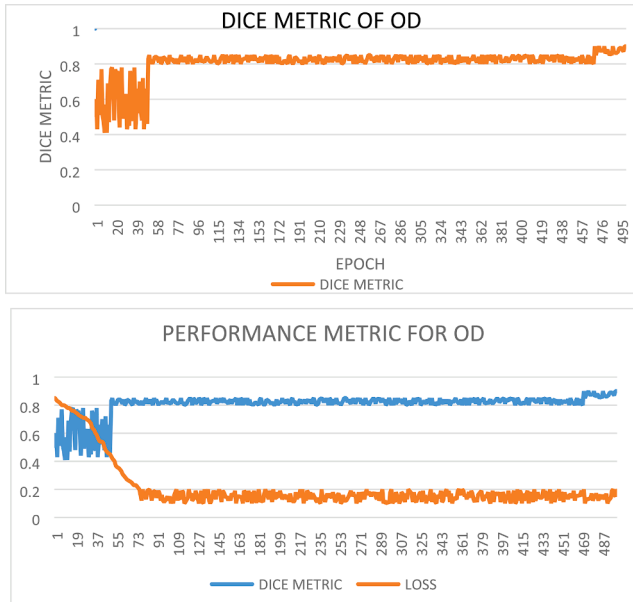


Fig. 22. Dice Metric and with its loss for OD segmentation and 500 epochs for trained data.

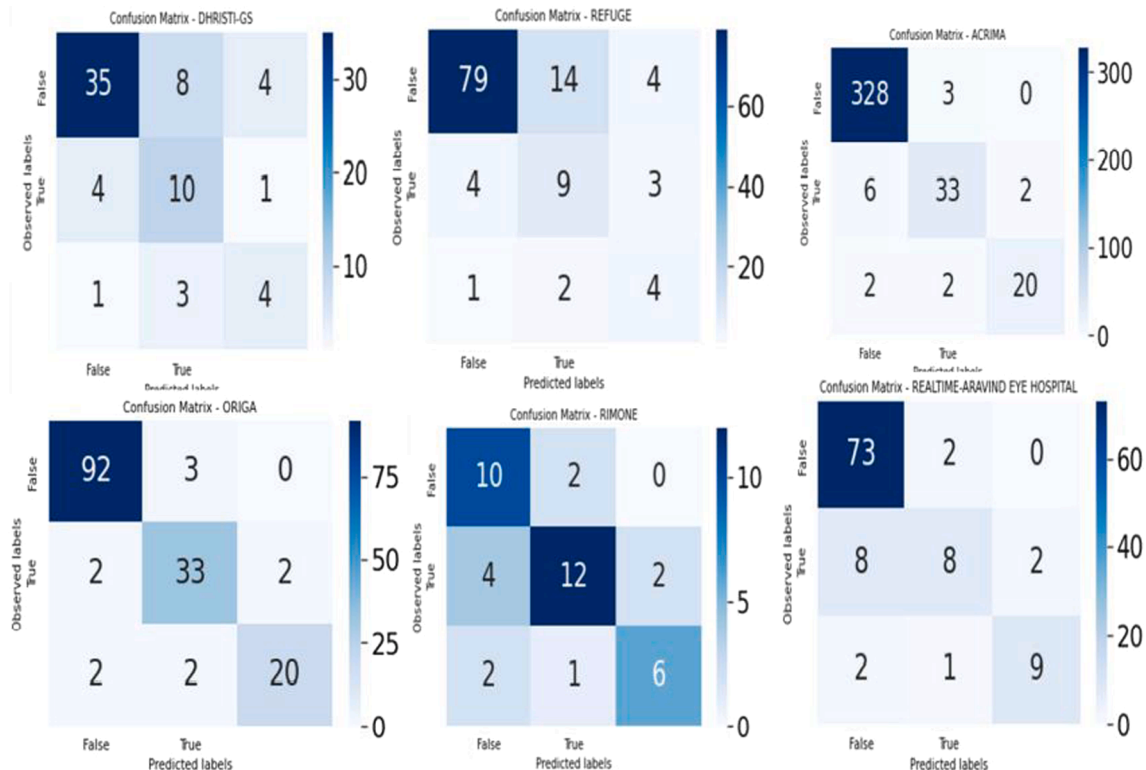


Fig. 23. Confusion matrix for every dataset for the model hybrid PolyNet.

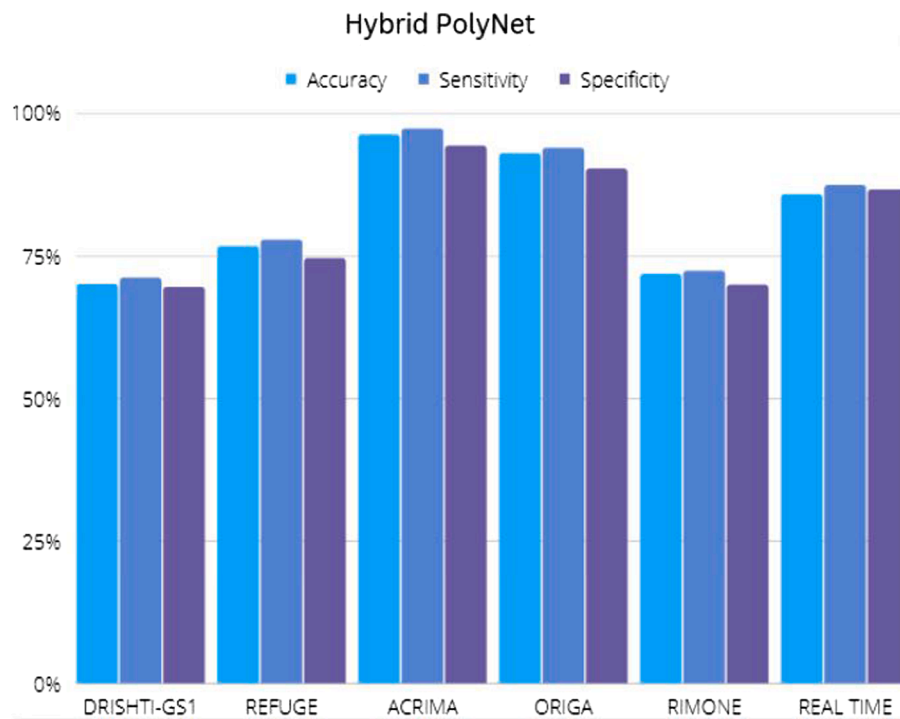


Fig. 24. Graph of the accuracy attained for every dataset.

Table 5

Comparison table of proposed dice metric of OD and OC with an existing work.

Related work	Epoch	Dice Metric (OD)	Dice Metric (OC)
Zilly et al.[20]	1000	0.85	0.83
Jiang et al. [7]	500	0.97	0.95
Juneja et al. [8]	500	0.93	0.92
Proposed work	500	0.98	0.97

Table 6

Training performance of the model on different epochs of the ACRIMA for the model hybrid PolyNet.

Epoch	Training Loss	Training Accuracy	Validation Loss	Validation Accuracy
1/500	2.0726	0.4900	2.1421	0.2421
100/500	1.2744	0.4400	1.3600	0.3431
200/500	0.6629	0.8233	0.8237	0.7605
300/500	0.2653	0.8632	0.3871	0.9148
400/500	0.1154	0.9591	0.1904	0.9621
500/500	0.0740	0.9621	0.0982	0.9621

glaucomatous images correctly and misinterpreted 6 images as mild and 2 images as severe. Additionally, 20 severe glaucomatous images were identified correctly and misinterpreted 2 images as mild and 2 images as moderate which attains the accuracy of 96.21 %.

Fig. 23 shows the confusion matrix for Drishti-GS1, REFUGE, ACRIMA, ORIGA, RIM-ONE and the real-time datasets performed by the proposed hybrid PolyNet. The severity of the class such as mild, moderate and severe levels of glaucoma has been performed for the datasets Drishti-GS1, REFUGE, ORIGA, RIM-ONE and the real-time datasets. The metrics such as accuracy, precision, recall and F1-score are evaluated in Table 7. The overall accuracy attained for the Drishti-GS1 dataset is about 70 %, the REFUGE dataset is about 83.67 %, the ACRIMA dataset

is about 96.21 %, the ORIGA dataset is about 92.95 %, the RIM-ONE dataset is about 71.79 % and the real-time dataset is about 85.71 %. Comparing to the other dataset, ACRIMA dataset achieves higher accuracy. Fig. 24 shows the graph for the accuracy attained for every dataset.

Table 8 shows the comparison of the proposed Hybrid PolyNet with the existing work. According to [31], [32], [33], [34] and [36] most of them have used the limited number of dataset, though attaining a better level of accuracy. In [36], Akter et al., used the ImageNet database with ResNet18 and VGG 16, has achieved 95.68 % and 96.2 % with very good results. But the ImageNet dataset deals with optical coherence tomography (OCT). The current work deals with the retinal fundus image and achieves the highest accuracy of 96.21 % in the ACRIMA dataset with sensitivity and specificity of about 97.32 % and 94.26 % respectively.

However, artificial intelligence structures are yet capable of replacing ophthalmologists in the diagnosis of retinal disorders. As AI systems are designed for specialized tasks, ophthalmologists are trained to diagnose a wide variety of eye diseases. Ophthalmologists would benefit from having a good knowledge of Deep learning thus providing increased care and attention to a greater number of patients. Concerning additional research, the paper intends to concentrate on enhancing sensitivity and accuracy by implementing more modern transfer learning technology as part of an upgraded methodology to achieve improved classification outcomes.

5. Conclusion

Glaucoma is the main disease that causes Ocular Nerve Disease, which leads to vision loss. Early detection and prevention is the only precaution to overcome vision blindness. Manual checking of glaucoma needs more time and it also needs several ophthalmologists to confirm the prediction of glaucoma. The main focus of this paper is to identify the severity of glaucoma disease. The retinal fundus image goes through pre-processing, including normalisation, filtration, and template matching. The unwanted noise is eliminated after pre-processing, and the resulting image is then used to carry out feature extraction. Segmenting blood vessels is accomplished with the proposed U-Net model.

Table 7

Metrics for every dataset attained for the proposed hybrid polyNet model.

Dataset Name	True Positive	Overall Accuracy	Class	Accuracy	Precision	Recall	F1-Score
DRISHTI-GS1	49	70 %	Mild	75.71 %	0.74	0.88	0.80
			Moderate	77.14 %	0.67	0.48	0.56
			Severe	87.14 %	0.50	0.44	0.47
REFUGE	92	83.67 %	Mild	80.83 %	0.81	0.94	0.87
			Moderate	80.83 %	0.56	0.36	0.44
			Severe	91.67 %	0.57	0.36	0.44
ACRIMA	381	96.21 %	Mild	97.22 %	0.99	0.98	0.98
			Moderate	96.72 %	0.80	0.87	0.84
			Severe	98.48 %	0.83	0.91	0.87
ORIGA	145	92.95 %	Mild	95.51 %	0.97	0.96	0.96
			Moderate	94.23 %	0.89	0.87	0.88
			Severe	96.15 %	0.83	0.91	0.87
RIMONE	28	71.79	Mild	79.79 %	0.83	0.63	0.71
			Moderate	76.92 %	0.67	0.80	0.73
			Severe	87.18 %	0.67	0.75	0.71
REAL TIME DATASET	90	85.71 %	Mild	88.57 %	0.97	0.88	0.92
			Moderate	87.62 %	0.44	0.73	0.55
			Severe	95.24 %	0.75	0.82	0.78

Table 8

Comparison of the proposed with the existing work.

Author	Model	Accuracy	Sensitivity	Specificity	Dataset
[31]	Deep CNN	79.04 %	74.30 %	70.03 %	REFUGE
[32]	RCNN	87.40 %	85 %	71.17 %	ORIGA, HRF, OCT & CFI
[33]	RCNN	91.51 %	92.33 %	90.90 %	Fundus images – 2554, glaucoma – 1023 non - glaucoma 1531
[34]	Deep CNN	95.45 %	96.43 %	93.75 %	Normal – 402, Early AMD – 583 Wet AMD – 125
[36]	ResNet18	95.68 %	100 %	95.6 %	ImageNet database
[36]	VGG16	96.2 %	100 %	95.6 %	ImageNet database
[35]	RF model	98 %	98.3 %	97.5 %	Normal – 202 healthy glaucoma – 297
Proposed work	Hybrid PolyNet	70 %	71.1 %	69.5 %	DRISHTI-GS1
		76.67 %	77.78 %	74.52 %	REFUGE
		96.21 %	97.32 %	94.26 %	ACRIMA
		92.95 %	93.90 %	90.25 %	ORIGA
		71.79 %	72.32 %	69.87 %	RIMONE
		85.71 %	87.33 %	86.55 %	REAL TIME DATASET

To effectively capture the optic cup during feature extraction, the optic disc must first be segmented. After the optic disc has been removed, the optic cup is split to locate the CDR, which determines the degree of glaucoma severity. Deep neural network models have been used to accurately predict the severity of glaucoma, including normal, mild, moderate, and severe levels including the classification of glaucoma or non-glaucoma. The segmentation of blood vessels is implemented using the proposed U-Net model, such that the sensitivity and the specificity are measured as 97 % and 88 % respectively. The accuracy and the F1 score are measured as 96 % and 98 % respectively. Hence, the proposed deep learning architecture with a hybrid PloyNet model is implemented for finding the severity level of glaucoma to produce an appropriate result. By increasing the number of layers, more features are identified.

Mainly the sampling process provides the exact classification of the severity level of glaucoma with an accuracy of 96.21 % in the ACRIMA dataset.

Future Enhancement

The proposed method is very effective in detecting the automated measurement of CDR in glaucoma at the earlier stage. The result from the proposed work has attained high accuracy compared to the existing work. But there may be a lot in improvement for the future is required. Artificial intelligence (AI) structures are not yet capable of replacing ophthalmologists in the diagnosis of retinal disorders. This is due to the fact that AI systems are designed for specialized tasks, but ophthalmologists are educated to diagnose a wide variety of eye diseases. Ophthalmologists would benefit from having a good knowledge of Deep learning so that they may provide increased care and attention to a greater number of patients. Concerning additional research, intends to concentrate on enhancing the sensitivity and accuracy by implementing more modern transfer learning technology as part of an upgraded methodology in order to achieve improved classification outcomes. If the total numbers of glaucoma images are increased, it attains a high level of accuracy. Glaucoma disease can be detected by identifying descriptors combination of features with a multiscale classifier to grade the disease by more images to reduce the computational time. Enhanced architectures can be used to detect the finer blood vessels so that the prediction of OD gets improved.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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